



The 66th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

732.ALLOGENEIC TRANSPLANTATION: DISEASE RESPONSE AND COMPARATIVE TREATMENT STUDIES

Recovery of the Intestinal Microbiome in Patients at Day +100 after Allogeneic Stem Cell Transplantation Is Defined By Microbial Metabolite Profiles and Linked to Long-Term Outcomes

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The intestinal microbiome influences outcomes of patients undergoing allogeneic stem cell transplantation (allo-SCT). Reduced bacterial diversity and levels of immuno-modulatory microbial metabolites such as butyrate and propionate (short-chain fatty acids (SCFAs)) are associated with poor survival and higher incidence of acute intestinal graft-versus-host disease (GvHD) (Payen et al. 2020, Thiele Orberg et al. 2024). While previous studies have focused on diversity and metabolites in the peri-engraftment period up to day +28, the microbiome's composition is dynamic over time. Persistent dysbiosis at day +100 has been associated with higher rates of chronic intestinal GvHD, and specific serum SCFAs appear protective (Markey et al. 2020). Given the growing cohort of long-term allo-SCT survivors and their comorbidities, such as chronic (c)GvHD or risk of relapse, the investigation of intestinal microbial metabolites at later time points is of interest.

In this longitudinal prospective study, we hypothesized that differences in intestinal metabolite levels would accompany differences in microbial recovery in the post-transplant period, which would be associated with clinical outcomes. We enrolled 109 patients undergoing allo-SCT at the Technical University Munich, of which 20 patients with 112 stool samples met the inclusion criteria of having at least one sample in the following periods: *PRE-TRANSPLANT* (TX, day -7), *PERI-ENGRAFTMENT* (day 0 - day + 49), *EARLY POST-TX* (day + 50 - day + 92) and *LATE POST-TX* (day $\geq$ +93). We analyzed the microbiome by 16S rRNA amplicon sequencing, microbial metabolites by targeted mass spectrometry, and microbial pathways by metagenomics. Patients were considered 'recovered' with an effective richness value above the 25th percentile, averaged from day +56 - day +100 to baseline day -7. We then tested for correlation between group classification, metabolite levels, and clinical outcomes.

All patients showed reduced bacterial diversity, SCFAs, and branched-chain fatty acids (BCFAs) after allo-SCT. We stratified patients into "recovery" (n=9) and "no recovery" (n=11). There was no significant association between recovery and recurrence of metabolite levels, and bacterial recovery showed a non-significant association with increased Overall Survival (OS, p=0.0648, Kaplan-Meier estimator). We attributed this discrepancy to altered microbial communities with different functional capacities in the *LATE POST-TX* period (Peled, Gomes et al. 2020) compared to *PRE-TX*. We used beta diversity to analyze which patients returned to their *PRE-TX* microbiome and stratified groups into "similar" (n=9) and "not similar" (n=11). Despite being in the recovery group, three patients (33%) showed no similarity. We hypothesize that functional readouts, such as metabolite levels or microbial pathways in the *LATE POST-TX* period, may better predict clinical outcomes. We analyzed samples from two long-term survivors from the recovery and two patients from the no-recovery group. We found that SCFAs (acetic -, propionic-, butyric acid) and BCFAs (isobutyric -, 2-methyl-butyric -, isovaleric acid), their associated pathways (PWY

5676, -7383, -5022) and superclass pathways (acetate, butanoate) were more abundant in the two patients of the recovery and low/undetectable in the two patients from the no recovery group.

Our study's findings underscore that recovery, as assessed by alpha diversity (via 16S), does not necessarily mean patients return to their baseline microbiome after allo-SCT. Consistently, "recovery" is only loosely associated with clinical outcomes. Instead, functional readouts such as microbial metabolite levels and microbial pathways may provide a more precise view. Therefore, we are currently validating if metabolite profiles at the *LATE POST-TX* period can predict OS and incidence of cGVHD in an independent cohort with paired stool samples from day -7 and day +100 in $n=40$ patients receiving allo-SCT at the University Hospital Regensburg. These data will be presented at ASH 2024.

This investigation of microbial and metabolite recovery in the post-transplant period provides a promising rationale for using metabolite profiles as predictive biomarkers of clinical outcome. These findings could have significant implications for future microbiome-modulating therapies, such as fecal microbiota transplantation, and could improve long-term outcomes.

Disclosures Schwarz: *Beigene*: Honoraria, Other: Travel Grant. **Neuhaus:** *HIPP*: Research Funding. **Wolff:** *Incyte*: Honoraria; *Novartis*: Honoraria, Research Funding; *Mallickrodt*: Honoraria; *Sanofi*: Honoraria; *Behring*: Honoraria; *Takeda*: Honoraria; *Neovii*: Honoraria. **Herr:** *Amgen*: Other: Travel Grant; *Janssen*: Other: Research Grant to Institution, Travel Grant, Research Funding; *Sobi*: Other: Travel Grant. **Heidegger:** *Roche*: Current Employment, Current equity holder in publicly-traded company; *Genentech*: Current Employment, Current equity holder in publicly-traded company. **Bassermann:** *BMS*, *Janssen*: Honoraria. **Weber:** *Else-Kröner-Fresenius Foundation*: Research Funding; *German Jose-Carreras Leukemia Foundation*: Research Funding; *German Research Foundation*: Research Funding.

<https://doi.org/10.1182/blood-2024-199035>